

Machine Learning-Enhanced Portfolio Optimization for Tailored Investment Strategies Across Diverse Risk Appetites

Minh Duc Nguyen ^{1)*} 

¹⁾ Department of Economic Information Systems, University of Economics, Hue University, Hue, Vietnam

¹⁾ nguyenminhduc@hueuni.edu.vn

Abstract

Background: Although researchers have increasingly explored the combination of machine learning based return forecasts with traditional portfolio construction, the discussion about how these predictive models reshape established methods is still developing. One prominent direction involves extending the classical mean variance approach so that it incorporates forward looking estimates, which is often referred to as the Mean Variance with Forecasting (MVF) framework. In parallel, approaches such as Risk Parity Portfolios (RPP) and Maximum Drawdown Portfolios (MDP) continue to gain attention because they represent different perspectives on risk management. However, despite this growing interest, there is still limited empirical evidence on how Support Vector Regression (SVR) and Random Forest (RF) forecasts affect performance within these three frameworks, and this gap is particularly evident in emerging markets.

Objective: This study examines how SVR and RF one day ahead return forecasts influence the risk adjusted performance, drawdown control, and diversification outcomes of the MVF, RPP, and MDP frameworks when applied to stocks in the VN-100 index between 2017 and 2024. The choice of these frameworks is intentional, as each reflects a different level of investor tolerance for risk. MVF tends to appeal to investors who place greater weight on potential returns, RPP seeks a more even distribution of risk which suits investors with a moderate stance, and MDP focuses on limiting losses, making it more suitable for investors who are highly cautious about downside risk.

Methods: Daily returns of VN 100 stocks were standardized and then used as inputs for the SVR and RF models. The models were tuned through a grid search on data from 2017 to 2021 and evaluated on the remaining period up to 2024. After generating the return forecasts, portfolios were constructed under the MVF, RPP, and MDP frameworks, and their performance was assessed using monthly excess returns, the information ratio, and total returns in comparison with the VN-100 index.

Results: The forecasts generated by SVR showed greater reliability than those obtained from the RF model, and this contributed to stronger risk adjusted performance when applied within the MVF framework. The MDP strategy, which places emphasis on limiting drawdowns, delivered solid protection against large losses, whereas the RPP approach produced more moderate returns along with improved consistency.

Conclusion: In the end, matching forecasting techniques and portfolio construction methods with an investor's risk preferences and view of the market is crucial, since overall performance is shaped by the interaction between predictive inputs and allocation rules. Looking ahead, future studies could investigate a wider range of forecasting models, incorporate transaction costs more explicitly, and explore adaptive forms of optimization that are capable of responding to changing market conditions.

Keywords: Machine Learning, Maximum Drawdown, Mean-Variance, Portfolio Optimization, Random Forest, Risk Parity, Support Vector Regression, VN-100 Index

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I. INTRODUCTION

Portfolio optimization has been a core theme in financial research for decades, dating back to the seminal contribution of Markowitz (1952) [1], who established a formal approach to balancing risk and return through what became known as the Mean Variance framework. Over time, further innovations, including the Black Litterman model [2], risk parity methods [3], and strategies that prioritize drawdown control [4], have expanded the set of tools available to investors and acknowledged the reality that attitudes toward risk differ widely across individuals and market environments. One of the continuing difficulties in this field is the challenge of producing accurate forecasts for returns and covariances, since financial markets are noisy and subject to structural changes [5]. In recent years, machine learning techniques have emerged as a promising tool for improving predictive accuracy in financial contexts.

* Corresponding author

Algorithms such as Random Forest (RF) [6] and Support Vector Regression (SVR) [7] can capture complex, non-linear relationships in historical data, while more advanced deep learning [8] offers the potential to uncover latent structures in large-scale time series. Empirical evidence suggests that even modest gains in forecasting performance may lead to measurable improvements in portfolio outcomes [9], [10]. Nonetheless, researchers and practitioners continue to explore and discuss how to effectively integrate machine learning driven predictions into distinct portfolio construction schemes that cater to varying investor preferences, ranging from aggressive return focused strategies to conservative drawdown-oriented ones [11].

Although there has been a notable rise in the use of machine learning techniques for financial forecasting and portfolio design, there is still only a partial understanding of how these predictive models interact with optimization approaches that represent different levels of investor tolerance for risk. Much of the existing literature has examined developed markets, and most contributions have relied on a single optimization method or a conventional approach. For example, Pinelis and Ruppert (2023) [12] proposed maximally predictable portfolios using RF and SVR, and Masuda (2024) [13] evaluated hybrid stock prediction systems within a mean variance setting. Other works have coupled forecasting with established optimizers, including hybrid ARIMA-LSTM methods [14], clustering integrated with mean variance allocation [15], applications of machine learning with the Omega ratio and mean variance approaches [16], and ensemble-based techniques for trading strategies [17]. Together, these studies demonstrate the growing potential of machine learning to improve both forecasting and allocation decisions, yet they remain centered on advanced financial markets and largely confined to traditional optimization designs. As a result, important questions remain regarding how predictive models perform across frameworks that serve different risk attitudes, and how these insights may extend to developing markets.

This study adds to the ongoing discussion by evaluating the performance of two commonly used machine learning forecasting techniques, SVR and RF, in combination with three portfolio optimization frameworks: Mean Variance with Forecasting (MVF), Risk Parity Portfolio (RPP), and Maximum Drawdown Portfolio (MDP). Consistent with earlier work, we interpret risk appetite as a multidimensional concept that includes the investor's intrinsic tolerance for volatility and expected returns, as well as the investor's perspective on market uncertainty, prevailing sentiment, and the potential for losses [18], [19], [20]. The three portfolio frameworks are selected with this interpretation in mind. MVF is most relevant for investors who seek to enhance returns while acknowledging and controlling volatility through the mean variance relationship. RPP offers a more balanced stance that may suit investors who emphasize steady diversification, and MDP is intended for investors who place a strong emphasis on safeguarding capital and limiting losses. Through this lens, we examine the trade-offs that arise when different forecasting capabilities are paired with contrasting portfolio philosophies. To ground the analysis in a practical setting, we incorporate the VN-100 index as a benchmark and evaluate outcomes using monthly excess returns. By comparing the forecasting approaches in a systematic manner, the study offers insight into how predictive accuracy and risk management preferences work together to shape portfolio performance.

The contributions of this research are threefold. First, we provide a systematic comparison of forecasting-optimization combinations that explicitly accommodate diverse investor risk appetites, a perspective that has been underexplored in prior work. Second, we incorporate a real-world benchmark, the VN-100 index, and focus on monthly excess returns to assess practical investment performance. Third, by evaluating how predictive accuracy interacts with portfolio construction under different risk preferences, our study offers actionable insights for both academics and practitioners in the field of financial decision-making.

The remainder of this paper is organized as follows. Section II reviews the key literature on portfolio optimization and machine learning-based forecasting. Section III then describes the data, preprocessing steps and methodological details of our machine learning forecasts and portfolio construction. Following that, Section IV presents experimental results, analyzing performance of stock return prediction and portfolio optimization. Finally, Section V offers concluding remarks on the implications of our findings and propose avenues for future research.

II. LITERATURE REVIEW

Markowitz's mean-variance model [1] provided a quantitative foundation for portfolio construction by weighing expected returns against variance and defining the efficient frontier, yet its dependence on historical estimates often led to extreme and unstable allocations when market conditions shifted. To address these limitations, Black and Litterman (1992) [2] introduced a Bayesian refinement that blends equilibrium returns implied by market capitalization weights with investor views, pulling allocations toward a consensus benchmark and improving robustness beyond the sample period. Extending this progression, MVF replaces static historical averages with forward looking return forecasts and their associated forecast errors to better align allocations with evolving market dynamics [16], [21], and empirical evidence shows that even modest gains in predictive accuracy can yield meaningful

improvements in risk adjusted outcomes. Risk Parity moves attention away from estimates of return by equalizing each asset's contribution to total volatility, which tends to produce smoother drawdowns and stronger diversification, although it may reduce exposure to assets with higher expected returns [3]. Roncalli (2014) [22] advanced this idea by scaling risk budgets in line with expected returns, preserving diversification and downside control while adding a forward-looking tilt. At the same time, strategies that emphasize drawdown management, including those based on Conditional Drawdown at Risk [4], explicitly target peak to trough losses across a chosen horizon, offering strong capital protection for investors who are especially averse to losses, albeit with greater computational demands. Collectively, these frameworks reflect distinct investor risk appetites, with MVF favored by return oriented investors, RPP suited to those seeking balanced diversification, and MDP aligned with conservative investors focused on minimizing losses. This alignment underscores the importance of evaluating forecasting models across multiple optimization approaches, since real world investment behavior varies widely and cannot be captured through a single framework.

Over the past two decades, machine learning methods have grown in popularity for financial forecasting due to their ability to uncover non-linear patterns in noisy time-series data [10], [11]. Techniques such as SVR and RF have repeatedly shown predictive advantages over simpler linear models, while deep learning algorithms expand the modeling frontier further [8]. RF has been effectively applied in financial return prediction, with Patel et al. (2015) [23] demonstrating that combining RF with trend-deterministic preprocessing and technical indicators yields significant improvements in directional accuracy and risk-adjusted returns. Zheng et al. (2024) [24] applied RF to U.S. stock market data and achieved superior accuracy and robustness during volatile periods. Moreover, recent heteroscedastic SVR models integrating autoregressive components have enhanced stability under non-stationary market conditions [25]. Readers are encouraged to explore the application of machine learning and deep learning techniques in financial forecasting through comprehensive survey papers [26], [27]. These insights align with studies by Gu et al. (2020) and Bianchi et al. (2021) [9], [10], which highlight that even small improvements in forecast accuracy can significantly enhance portfolio returns when effectively integrated into a suitable allocation framework.

Recent studies have begun to combine machine learning based forecasts with portfolio optimization in an effort to enhance returns while acknowledging practical constraints. Ma et al. (2021) [16], for example, employ RF and SVR to generate return predictions and integrate them into mean variance and Omega based portfolios. Their assessment explicitly accounts for transaction costs and uses a custom index as a benchmark, showing that improvements in forecasting accuracy can still translate into net performance gains after trading expenses are considered. In a similar vein, Behera and Kumar (2025) [17] explore the use of time series models, such as ARIMA, together with machine learning to construct portfolios within a mean variance framework, and they introduce their own benchmark to evaluate whether the resulting allocations outperform alternative approaches once transaction costs are included. Other related research includes the maximally predictable portfolios developed by Pinelis and Ruppert (2023) [12] using RF and SVR, and the hybrid stock prediction models examined by Masuda (2024) [13] within a mean variance setting. Additional contributions involve hybrid ARIMA-LSTM models [14] and clustering combined with mean variance optimization [15], further illustrating the growing interest in integrating forecasting tools with established portfolio methods.

However, despite these recent advances, much of the existing literature remains confined to a single optimization approach or focused on a specific market environment, and there has been limited examination of how different risk management frameworks respond to improved forecasts in emerging markets. In particular, the Vietnamese equity market, characterized by variable liquidity, regulatory nuances and unique market microstructure, has received limited systematic study in this domain. By evaluating SVR- and RF-powered MVF, RPP and MDP strategies on the VN-100 index, our work addresses this gap and provides comprehensive insights into the interplay between predictive model accuracy and portfolio construction methods in an emerging-market context.

III. METHODS

This section presents the experimental design including data preparation, predictive modeling and portfolio optimization frameworks. The design begins with an overview of the data preprocessing steps. Next, two machine learning models (SVR, RF) are introduced. Finally, three portfolio optimization frameworks are discussed including MVF, RPP and MDP, each designed to cater to different risk management approaches based on the forecasts provided by the machine learning models.

A. Data and preprocessing

The dataset used in this study is derived from the VN-100 index, a prominent benchmark in the Vietnamese stock market that tracks the performance of the top 100 companies listed on major exchanges in Vietnam. The VN-100 serves as a critical gauge of market performance and investor sentiment in one of Asia's emerging markets. For this

analysis, we extracted the daily closing prices of stocks within the VN-100 index covering the period from 2017 to 2024 using the Python `vnstock` library. To ensure data integrity, we excluded stocks that were not continuously listed during the study period or that contained missing data. After applying these criteria, the final dataset consists of 59 stocks, providing a robust foundation for our predictive analysis. To capture recent market dynamics, we employ a sliding window approach that uses the last 30 days of daily returns as the input for our prediction models. This method not only emphasizes short-term trends but also maintains a manageable input size for the machine learning algorithms deployed in this study. Prior to model training, the data is standardized to enhance convergence and overall performance. Standardization is achieved by transforming each feature so that it has a zero mean and unit variance. Specifically, for each original feature value x , the standardized value z is computed using the equation:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where μ is the mean and σ is the standard deviation of the feature. This transformation ensures that each feature contributes equally during model training by scaling them to a common range.

To preserve the temporal integrity of the time series data and avoid look-ahead bias, the dataset is partitioned into distinct chronological segments. The training set, spanning from 2017 to 2021, is used for model development and parameter tuning. The validation set, comprising data from 2022, facilitates fine-tuning and helps prevent overfitting. Finally, the test set, which includes data from 2023 to 2024, is employed to evaluate the predictive performance on unseen data. This sequential split mirrors real-world forecasting conditions and provides a comprehensive framework for assessing the robustness of the models.

B. Prediction models

1) Support Vector Regression (SVR)

SVR is a robust supervised learning algorithm widely employed for regression tasks, including financial forecasting [7]. Originating from the principles of SVM, SVR uses margin optimization to determine a function that approximates the target values while tolerating deviations within a specified threshold. By incorporating kernel functions, SVR effectively maps input data into a higher-dimensional space, enabling the capture of complex, non-linear relationships that are particularly prevalent in stock price movements.

In this study, we utilize SVR with the Radial Basis Function (RBF) kernel. The RBF kernel is well-suited for modeling non-linear patterns in financial time series data due to its ability to measure the similarity between data points in the transformed feature space. The RBF kernel function is defined as:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (2)$$

where x_i and x_j are input vectors and γ is a hyperparameter that controls the influence of individual training examples. A smaller γ value implies a wider similarity region, while a larger γ value results in a narrower region, making the model more sensitive to individual data points. The performance of the SVR model is highly dependent on the appropriate selection of hyperparameters. Among the most important are the regularization parameter C , which balances model complexity against training error; the kernel coefficient γ , which determines the reach of the RBF kernel; and the margin parameter ε , which specifies the width of the margin within which errors are not penalized.

To identify the optimal combination of these hyperparameters, we performed a grid search using cross-validation on the training set. Table 1 summarizes the range of values tested during this process.

Hyperparameter	Values tested
C	1, 2, 4, 8, 16, 25
γ	2^{-4} , 2^{-3} , 2^{-2} , 2^{-1} , 2^0
ε	2^{-4} , 2^{-3} , 2^{-2} , 2^{-1} , 2^0

2) Random Forest (RF)

RF is an ensemble learning method that constructs multiple decision trees and aggregates their predictions to improve accuracy and reduce overfitting [6]. In each tree, a subset of the data is sampled (with replacement) to create a bootstrap sample and a random subset of features is considered when splitting each node. This dual-randomization approach helps mitigate correlation among trees and enhances the model's generalization capabilities.

Mathematically, in a RF regressor composed of N trees, the overall prediction $\hat{y}$ for a given input x is obtained by averaging the individual predictions from each decision tree:

$$\hat{y}(x) = \frac{1}{N} \sum_{i=1}^N f_i(x) \quad (3)$$

where $f_i(x)$ denotes the prediction from the i -th tree. In this study, we fix the number of decision trees, N , at 500, following the recommendation by Breiman (2001) [6]. Although the optimal number of trees can vary depending on the dataset, 500 is typically sufficient to stabilize predictions without incurring prohibitive computational costs.

Similar to our approach with SVR, a grid search with cross-validation is employed to fine-tune additional hyperparameters. Table 2 summarizes the range of hyperparameter values evaluated during the grid search. Common parameters that can significantly impact RF performance include `max_depth`, which controls the maximum depth of each tree; `min_samples_split`, which specifies the minimum number of samples required to split an internal node; `min_samples_leaf`, which defines the minimum number of samples required at a leaf node; and `max_features`, which determines the number or proportion of features to consider when searching for the best split.

TABLE 2
GRID SEARCH RANGES FOR RF HYPERPARAMETERS

Hyperparameter	Values tested
<code>max_depth</code>	10, 15, 20, 25, 30
<code>min_samples_split</code>	5, 10, 15, 20, 25
<code>min_samples_leaf</code>	5, 10, 15, 20, 25
<code>max_features</code>	10, 20, 30, 40, 50

The configurations yielding the best validation performance were used to train the final models for both SVR and RF. This systematic hyperparameter tuning process involved evaluating various hyperparameter combinations through cross-validation on the training dataset. By selecting the configuration with the best validation performance, the process ensures that both models are finely calibrated to capture the complex relationships in stock market data. This approach balances model complexity, computational efficiency and predictive accuracy, enhancing each model's ability to generalize to unseen time-series data.

C. Portfolio optimization frameworks

1) Mean-Variance with Forecasting (MVF)

MVF extends the classic Markowitz mean-variance framework [28] by integrating forward-looking predictions of expected returns rather than relying solely on historical averages, proposed by [21]. In a standard mean-variance setting, the goal is to balance variance (a proxy for risk) against expected returns. MVF modifies this traditional approach by replacing the historical mean of each asset's return with forecasts generated by machine learning models. The resulting optimization problem typically takes the form:

$$\min_w (w^T \Sigma w - \hat{\mu}^T w - \bar{\varepsilon}^T w) \quad (4)$$

subject to:

$$\sum_{i=1}^n w_i = 1, 0 \leq w_i \leq 1 (i = 1, \dots, n) \quad (5)$$

where $w = (w_1, \dots, w_n)^T$ is the vector of asset weights, Σ is the $n \times n$ covariance matrix of returns, $\hat{\mu}$ is the vector of predicted returns and $\bar{\varepsilon}$ is the vector of forecast-reliability scores. By incorporating these forecasts, MVF potentially aligns portfolio allocations more closely with near-term market dynamics than historical-mean-based methods.

2) Risk Parity Portfolio (RPP)

The traditional RPP, introduced by Edward Qian (2005) [3], seeks to equalize the contribution of each asset to the total portfolio risk, typically measured via volatility. However, this approach does not incorporate return expectations, potentially overlooking assets with superior performance potential. To address this, Roncalli (2014) [22] proposed a generalized risk budgeting framework that integrates return expectations into the optimization process. In his formulation, excess returns (i.e., returns above the risk-free rate) are used to define a risk measure that blends return and volatility. However, in this study, we adopt a simplified yet practical version by directly using expected returns

instead of excess returns. This allows us to incorporate predictive information without relying on a risk-free benchmark. Accordingly, the portfolio is obtained by solving the following optimization problem:

$$\min_w \left(-w^\top \mu + c \cdot \sqrt{w^\top \Sigma w} \right) \quad (6)$$

subject to:

$$\sum_{i=1}^n w_i = 1, 0 \leq w_i \leq 1 (i = 1, \dots, n) \quad (7)$$

where μ is the vector of expected returns, Σ is the covariance matrix of asset returns, w is the vector of portfolio weights, and $c > 0$ is a risk–return trade-off parameter controlling the balance between return maximization and risk minimization.

This objective function represents a standard deviation-based risk measure that encourages allocation to assets with higher expected returns while penalizing portfolio volatility. The parameter c governs the strength of risk aversion: lower values favor return-seeking behavior, while larger values approximate classic risk parity.

3) Maximum Drawdown Portfolio (MDP)

MDP focuses on minimizing the worst peak-to-trough loss within a specified time horizon [4]. For a portfolio with value P_t at time t , the drawdown D_t is given by:

$$D_t = \frac{\max_{\tau \leq t} P_\tau - P_t}{\max_{\tau \leq t} P_\tau} \quad (8)$$

and the maximum drawdown (MD) is the supremum of D_t over t . By seeking to minimize MD, the portfolio aims to protect capital against severe downturns. This approach often resonates with conservative investors who prioritize loss avoidance over capturing higher returns. The corresponding optimization problem is formulated as:

$$\min_w \max_{t \in [0, T]} \left(\frac{\max_{\tau \leq t} P_\tau - P_t}{\max_{\tau \leq t} P_\tau} \right) \quad (9)$$

subject to:

$$\sum_{i=1}^n w_i = 1, 0 \leq w_i \leq 1 (i = 1, \dots, n) \quad (10)$$

where $w = (w_1, w_2, \dots, w_n)^\top$ is the vector of portfolio weights, P_t is the portfolio value at time t , computed recursively as $P_t = P_{t-1}(1 + w^\top R_t)$, $R_t \in \mathbb{R}^n$ is the vector of asset returns at time t , and T is the investment horizon.

Taken together, MVE, RPP and MDP highlight varying degrees of risk and return emphasis. MVE targets higher returns with moderate to aggressive risk tolerance, RPP equalizes risk exposures to achieve diversification benefits and MDP prioritizes capital preservation by limiting large losses.

IV. RESULTS

A. Performance of stock return prediction

Evaluating the accuracy of predictive models is essential in financial forecasting, where small errors can lead to significant investment decisions. In this study, we assess the performance of our stock return predictions using two common regression metrics: Mean Absolute Error (MAE) and Mean Squared Error (MSE). The MAE measures the average magnitude of errors in the predictions, without considering their direction. It is defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

where y_i is the actual return, $\hat{y}_i$ denotes the predicted return and n is the total number of observations. Lower MAE values indicate that, on average, the model's predictions are closer to the actual values.

The MSE is another widely used metric in regression, placing greater emphasis on larger errors by squaring them. It is computed as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (12)$$

Similar to MAE, lower MSE values imply more accurate forecasts. However, because errors are squared, MSE penalizes larger deviations more heavily than MAE does. Table 3 presents the MAE and MSE values for the two forecasting models, SVR and RF, across the test set. These values reflect the models' ability to predict daily returns for the stocks in our dataset.

TABLE 3
THE PREDICTIVE PERFORMANCE OF SVR AND RF

Model		MAE	MSE
SVR	Mean	$1.50 * 10^{-2}$	$4.42 * 10^{-4}$
	Std	$3.76 * 10^{-3}$	$1.93 * 10^{-4}$
RF	Mean	$1.97 * 10^{-2}$	$8.04 * 10^{-4}$
	Std	$6.16 * 10^{-3}$	$4.33 * 10^{-4}$

As shown in Table 3, SVR exhibits lower average errors on both metrics compared to RF, indicating that SVR's predictions are more accurate on average. The significantly smaller standard deviations for SVR also suggest more consistent performance across different prediction intervals. Overall, these results imply that SVR may be better suited than RF for forecasting under the conditions tested.

B. Performance of portfolio optimization

This section evaluates the performance of six distinct portfolio strategies that result from pairing two forecasting models, SVR and RF, with three portfolio optimization frameworks: MVF, RPP and MDP. We assume no transaction fees and measure each strategy's monthly excess return relative to the VN-100 index over the test period from 2023 to 2024. Specifically, we compare the mean monthly excess return, the standard deviation of monthly excess return and the total return of each portfolio to gauge how effectively it balances risk and return in comparison to this benchmark. By using the VN-100 as a reference, we can contextualize each strategy's performance against a prominent market measure, thereby offering deeper insight into the relative benefits of different forecasting and optimization approaches. Table 4 compares six portfolio strategies formed by pairing the two forecasting models with the three portfolio optimization frameworks.

TABLE 4
THE PORTFOLIO OPTIMIZATION PERFORMANCE OF DIFFERENT STRATEGIES

Strategy	Monthly excess return		Information ratio	Total return (%)
	Mean (%)	Std		
SVR-MVF	2.230	0.039	0.570	51.290
RF-MVF	1.569	0.050	0.317	36.081
SVR-RPP	1.518	0.055	0.276	34.909
RF-RPP	1.458	0.055	0.264	33.530
SVR-MDP	1.912	0.046	0.414	43.977
RF-MDP	1.660	0.047	0.353	38.177

Among these strategies, SVR-MVF stands out with the highest mean monthly excess return (2.230%) and the lowest standard deviation (0.039), yielding the highest information ratio (0.570) and total return (51.290%). Meanwhile, the MDP-based strategies (SVR-MDP, RF-MDP) display moderate volatility and returns, offering a mid-range risk-return trade-off and focusing on limiting substantial drawdowns. By contrast, the RPP portfolios produce the lowest returns and risk-adjusted metrics, indicating that in this particular setting, balancing risk contributions did not leverage the machine learning forecasts as effectively as MVF or MDP did. Across all optimization frameworks, SVR forecasts consistently outperform RF in terms of both return and risk-adjusted performance. In particular, SVR-MVF emerges as the strongest option overall, underscoring the advantage of combining more precise predictive modeling with a mean-variance approach for investors willing to accept somewhat higher volatility in pursuit of superior returns.

In addition to numerical metrics, Fig. 1 illustrates the net value evolution of each portfolio strategies over the test period. The initial value of each portfolio is normalized to 1.0 at the start of the period, enabling a direct comparison of growth paths over time.



Fig. 1. Net value of different strategies.

As shown in Fig. 1, SVR-MVF finishes with the highest terminal value, followed by SVR-MDP in second place, while the remaining four strategies converge to similar end points. This ordering aligns with the total return ranking presented in TABLE 4, where SVR-MVF and SVR-MDP also occupy the top two spots. Overall, the chart illustrates how different forecasting models and optimization frameworks produce distinct performance profiles. Investors looking for higher returns might find SVR-MVF attractive, albeit with greater sensitivity to market swings, whereas more risk-averse investors may gravitate toward portfolios like SVR-MDP or RF-MDP for more consistent drawdown control. The RPP strategies (especially RF-RPP) strike a moderate balance but, in this study, do not reach the peak returns of MVF.

To analyze monthly performance in greater detail, Fig. 2 and Fig. 3 present the monthly excess returns generated by each strategy in 2023 and 2024, respectively. As identified, SVR-MVF demonstrates the best overall performance. During market downturns, SVR-MVF effectively limits losses, recording the lowest losses in months such as September and October 2023, April and June 2024. However, in bullish months (e.g., July 2023 and May 2024), SVR-MVF does not outperform other strategies, suggesting that it is less suited to capitalize on market upswings. In second place, SVR-MDP performs exceptionally well in certain months, such as April 2023 and February 2024. However, it also suffers significant losses in March and June 2024, reflecting high volatility in its returns. Other strategies show moderate performance overall, with occasional strong months but generally higher volatility and less effective loss control compared to SVR-MVF.

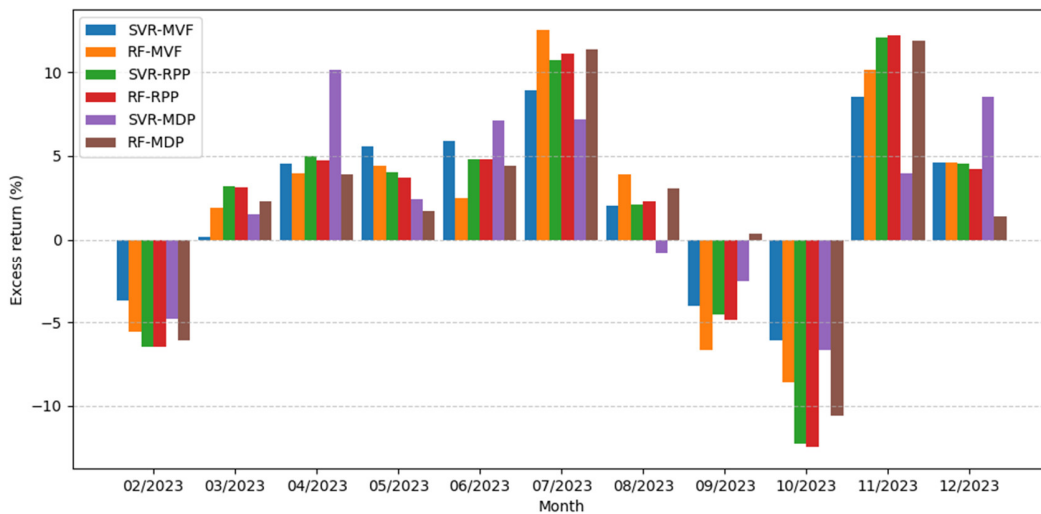


Fig. 2. Excess returns from various strategies during the year 2023

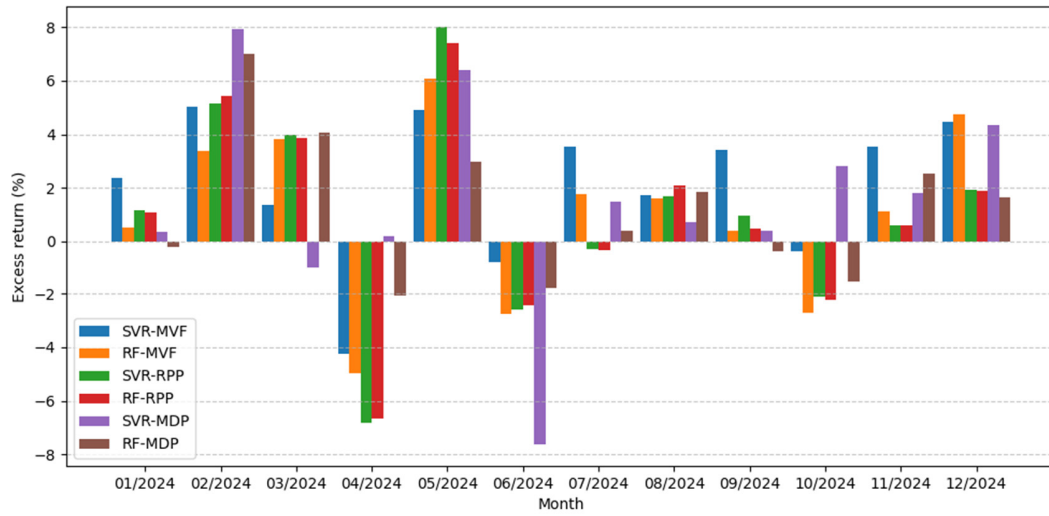


Fig. 3. Excess returns from various strategies during the year 2024

In summary, the analysis evaluated key performance metrics derived from monthly excess returns relative to the VN-100 Index. It also assessed net value over the test period and examined how each strategy's monthly excess returns reflected its response to different market conditions. This approach highlights the strengths, weaknesses and adaptability of each strategy across varying market environments.

V. DISCUSSION

The empirical results of this study offer insight into the way machine learning forecasting models interact with different portfolio optimization approaches when investor risk preferences vary. Both SVR and RF demonstrate solid predictive power for short term return forecasts, as indicated by relatively low MAE and MSE values across the VN 100 dataset. SVR achieves the lowest MAE and MSE, suggesting that it provides more accurate estimates of short-term stock returns. RF, in contrast, generates higher prediction errors and displays greater variability, which points to reduced stability from one run to another. Although ensemble methods such as RF are often viewed as resilient to noise and non-linear patterns, SVR performs better in this context, likely because its kernel-based structure can capture complex relationships when parameters are well tuned.

When these predictive models are incorporated into portfolio construction, the resulting strategies display noticeably different risk return profiles. The MVF approach records the highest average return, which aligns with its emphasis on balancing expected return and volatility. Yet this benefit is accompanied by higher volatility, meaning that it is best suited for investors who can tolerate greater risk. The RPP framework delivers a steadier performance by equalizing risk contributions across assets, offering a reasonable compromise between return and stability. The MDP approach produces the lowest return but provides strong protection against losses, which underscores its relevance for investors with a cautious outlook. Taken together, these findings demonstrate the importance of selecting forecasting techniques and optimization frameworks that match the investor's risk appetite. In addition, Table 5 compares the methods and outcomes of earlier studies with those of the present work, highlighting differences in research focus and practical implications.

TABLE 5
COMPARISON OF PREDICTION AND OPTIMIZATION MODELS ACROSS STUDIES.

Study	Prediction model	Optimization method	Dataset	Best prediction model	Best optimization model
Ma et al. (2021) [16]	SVR, RF, DMLP, LSTM, CNN	MVF, OF	Stocks from China Securities 100 Index	RF	CNN-MVF and LSTM-MVF
Ashrafzadeh et al. (2025) [15]	LSTM, CNN, RF, XGBoost	MVF	Seventeen large-cap stocks from the New York Stock Exchange	K-Means-LSTM	K-Means-LSTM-MVF
Our study	SVR, RF	MVF, RPP, MDP	Stocks from VN-100 Index	SVR	SVR-MVF

Note: DMLP = Deep multilayer perceptron; LSTM = Long short term memory; CNN = Convolutional neural network; XGBoost = Extreme Gradient Boosting; OF = Omega with forecasting.

Specifically, Ma et al. (2021) [16] tested a broad range of predictive models, where RF achieved the most accurate forecasts due to its ensemble averaging mechanism, which mitigated overfitting in the relatively stable China Securities 100 Index. Ashrafzadeh et al. (2025) [15] focused on diversification through clustering-based LSTM forecasting, where the K-Means-LSTM-MVF combination yielded superior performance. This improvement occurred because clustering reduced inter-asset correlation, allowing the MVF framework to allocate weights more efficiently. In contrast, our study employs SVR and RF predictors with three complementary optimization methods (MVF, RPP, and MDP) applied to the more volatile VN-100 Index. The SVR-MVF model delivered the highest performance because SVR effectively captured nonlinear relationships and was less sensitive to noise, while MVF dynamically adjusted portfolio weights according to predicted returns and risk levels. These results suggest that while previous studies achieved improvements through enhanced prediction accuracy or diversification, our framework's advantage stems from integrating multiple optimization perspectives to accommodate different investor risk appetites and market volatility conditions.

From a broader perspective, the combination of machine learning forecasting and portfolio optimization underscores the growing importance of data-driven decision-making in emerging markets. Unlike studies focusing on developed markets, this research provides evidence from Vietnam, where market dynamics, liquidity conditions, and investor behaviors differ significantly. The results suggest that while advanced forecasting methods can enhance portfolio construction, contextual adaptation remains critical for their effective application in developing financial markets.

Despite these promising outcomes, several limitations and threats to validity should be acknowledged. First, the study relies solely on historical price data, excluding other potentially influential variables such as macroeconomic indicators, sentiment data, or firm-specific fundamentals. Incorporating these factors could improve predictive accuracy and robustness. Second, the analysis focuses on a specific market (Vietnam), which may limit the generalizability of the findings to other emerging or developed markets. Third, the models assume frictionless trading and do not account for transaction costs or liquidity constraints, which could affect real-world portfolio performance. Additionally, the sliding window approach emphasizes short-term prediction horizons, which might not capture long-term structural shifts or regime changes. Finally, although model evaluation followed a chronological validation framework to prevent look-ahead bias, unforeseen data dependencies could still influence results.

VI. CONCLUSIONS

This study examined the interaction between two machine learning forecasting models, SVR and RF, and three portfolio optimization frameworks: MVF, RPP and MDP. The analysis evaluated both forecasting accuracy, where SVR demonstrated lower prediction errors (in terms of MAE and MSE) than RF and portfolio performance, where SVR-based strategies outperformed RF-based strategies in risk-return metrics and monthly excess returns relative to the VN-100 index. With its stronger predictive performance, the SVR-MVF strategy emerged as the best performer in portfolio outcomes, illustrating the benefits of accurate return forecasting for enhancing risk-return efficiency. The integration of SVR forecasts into the mean-variance framework enabled the strategy to capitalize on bullish periods while maintaining manageable drawdown levels. MDP-based strategies displayed a balanced profile, providing reliable drawdown protection but limited upside potential. In contrast, RPP-based portfolios exhibited subdued returns, reflecting the trade-off between balanced risk contributions and the ability to fully capture market gains.

The study contributes to the growing literature on data-driven investment strategies by demonstrating that forecasting precision and optimization design jointly shape portfolio outcomes, especially in emerging markets, where volatility and information asymmetry remain high. It also provides empirical evidence that different optimization frameworks correspond to distinct investor profiles: MVF suits return-oriented investors, RPP balances diversification and stability, and MDP offers effective downside protection. Future research could extend this framework by incorporating transaction costs, dynamic rebalancing, and alternative forecasting models to further evaluate the robustness of machine learning-assisted portfolio strategies under changing market conditions.

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Animal Subjects: There were no animal subjects.

ORCID:

Minh Duc Nguyen: <https://orcid.org/0000-0003-4702-0888>

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